SKITII TRIALS PLATFORM

15-Day Task List

Day-by-day delivery plan for Database Schema, Mobile App Backend APIs, and Admin Panel Backend APIs (Days 1–15 of 32).

VERSION
1.0

DATE
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CUSTOMER
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SCOPE
Trial Backend (Mobile App APIs + Admin Panel APIs) + Database Schema

Goal for Day 15

By end of Day 15:

- Database schema complete
- All **Mobile App Backend APIs** complete
- All **Admin Panel Backend APIs** complete
- Ready for Admin Panel Frontend (starts Day 12, continues after Day 15)

Overview

Days	Workstream	Duration
1 – 3	Database Schema Design	3 days
3 – 12	Backend APIs — Mobile App	9 days
8 – 15	Backend APIs — Admin Panel (parallel)	8 days

> Admin Panel Backend starts on Day 8 and runs in parallel with Mobile App APIs.

Days 1 – 3 — Database Schema Design

Day	Tasks	Status
Day 1	Design & implement core collections: `trialConfiguration`, `patients`, `sessions` — define indexes, field types, `institutionId` / `trialId` scoping	☐
Day 2	Design & implement form collections: `preTherapy`, `postTherapy`, `weeklyForm`, `weeklyFormPin`, `missedSessionLog`, `trialStatistics`	☐
Day 3	Finalize schema — add DB indexes, team review. Update existing Skitii DB: add `controlPatientMusic` flag to `musicAssets`. Begin Mobile App Backend APIs.	☐

Deliverable: All MongoDB collections ready for API development.

Days 3 – 12 — Backend APIs (Mobile App)

Day	Tasks	Status
Day 3	Device PIN APIs — create PIN, verify PIN, forgot PIN / OTP reset	☐
Day 4	Patient APIs — enroll (duplicate Skitii ID validation, dynamic schema), list, view, update (Week 0 lock)	☐
Day 5	`POST /sessions/start` — combined validation: shift window, shift ordering, duplicate session (same + different device), 30-min minimum check	☐
Day 6	`POST /sessions/validate` (safety-net after Pre Therapy), session stop (`INCOMPLETE` / `USER_ENDED`) / complete (`COMPLETED` / `SESSION_MAX_REACHED`) / readings upload, 20-min auto-end + 30-min timeout job (`AUTO_TERMINATED` / `TIMEOUT`)	☐
Day 7	Pre Therapy form API, Post Therapy form API, `TRIAL_COMPLETED` enforcement	☐
Day 8	Weekly Form PIN — `POST /weekly-assessments/verify-pin` (lockout, 1-hr token). Steps 1–4 batched submit with PIN token. **Admin Panel Backend starts.**	☐
Day 9	Weekly form Step 5 HRV submit, Week 0 gate enforcement, sequential form ordering rules	☐
Day 10	Statistics API with mandatory date range filter, shift window + shift ordering enforcement	☐
Day 11	Cron jobs — Sunday night PIN generation, 4 AM daily missed session email (admin + institute), Thu/Fri/Sat weekly form reminder email	☐
Day 12	Mobile App API polish — edge cases, error codes, `trialConfiguration` fetch API	☐

Deliverable: All Mobile App Backend APIs ready for React Native integration.

Days 8 – 15 — Backend APIs (Admin Panel)

Runs in parallel with Mobile App APIs from Day 8.

Day	Tasks	Status
Day 8	`trialConfiguration` CRUD — create / update trial config per institute (shift windows, patient schema, group types)	☐
Day 9	Patient admin APIs — full list, detail view, export	☐
Day 10	Session admin APIs — session history and readings per patient	☐
Day 11	Form admin APIs — Pre Therapy, Post Therapy, Weekly Form (all 5 steps) view per patient	☐
Day 12	Statistics admin API — detailed analytics, CSV / PDF export. **Admin Panel Frontend starts.**	☐
Day 13	Institute management APIs — add / manage hospitals, trial assignments	☐
Day 14	Music assets API — flag / unflag `controlPatientMusic`; verify `GET /music/control`	☐
Day 15	Admin Panel backend review, integration testing with mobile APIs, resolve conflicts	☐

Deliverable: All Admin Panel Backend APIs complete and ready for frontend integration.

Parallel Work View (Days 1–15)



What Comes After Day 15 (Reference)

Days	Work
12 – 18	Trial Admin Panel Frontend
12 – 14	Skitii Admin Panel Update (`controlPatientMusic` flag)
19 – 23	Deployment (UAT + Production)
22 – 32	Testing & Bug Fixes

Full project duration: 32 working days.

Checkpoint Summary for Management

Checkpoint	Day	Expected Outcome
Schema ready	Day 3	All collections designed and implemented
Core session APIs ready	Day 7	Patient + Session start/stop/complete + forms working
Weekly + cron ready	Day 11	Weekly forms, PIN, and all cron jobs working
Mobile APIs complete	Day 12	Full mobile backend ready for app team
Admin APIs complete	Day 15	Full admin backend ready for frontend

Risks for First 15 Days

Risk	Mitigation
Dynamic patient schema complexity	Finalize schema config format before Day 4 (Patient APIs)
Multi-tablet session conflict edge cases	Unit tests during Days 5–6 before mobile integration
Cron / email delivery issues	Test email templates early on Day 11
Parallel Admin + Mobile API conflicts	Shared service layer review on Day 15